

2. CLASSIFICATION 1

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- **Classification** is the grouping of living organisms based on their structure.
 - Those with similar structures are put under one group called **taxon (plural- taxa)**
 - A science/study of classification is called **taxonomy** and scientists are called **taxonomists**.
 - Taxonomy involves the placing of organisms into taxa and assigning them names.
- External features are used when classifying organisms e.g.
 - a) In plants-
 - i. Rhizoids and seta in moss plant.
 - ii. Fronds, sori and rachis in ferns.
 - iii. Roots, stems, leaves, flowers, seeds, fruits and cones in higher plants.

- b) In animals-
- i. Tentacles in hydra.
 - ii. Feathers and wings in birds.
 - iii. Shells in snails.
 - iv. Sensory organs e.g. eyes, ears, antennae.
 - v. Fur, hair and mammary glands in mammals.
 - vi. Scales and fins in fish.
 - vii. Proglotids and scolex in tapeworms.
 - viii. Locomotory structures e.g. limbs in arthropods and vertebrates.
 - ix. Body pigmentation.

Significance/ importance / necessity of classification.

1. It groups together living organisms with similar characteristics but separates those with different features.
2. Helps in placing living organisms into their correct groups for easy reference.
3. Helps to arrange the information about living organisms in an orderly manner to avoid chaos and confusion.
4. Helps to understand the evolutionary relationships between different organisms.
5. It makes it easier for scientists to communicate since the whole world uses the same groupings.

- Some organisms are small or have tiny features which cannot be seen clearly, therefore require the use of magnifying instruments
- A common magnifying instrument used is called a **hand lens**.
- It is used to magnify organisms/ make them look bigger.
- It is made up of a **convex lens** mounted on a **frame**.
- **The handle** is either wooden or plastic.

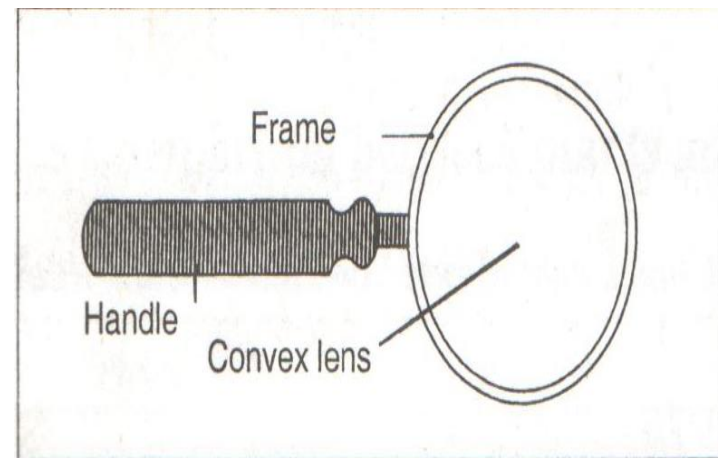


Fig. 2.1 A hand lens

How to use a hand lens.

- i. Place the object on the bench near the light source.
- ii. Move the lens to and from the eyes until the object becomes clear. When the object is clear, it is said to be in focus and an enlarged image is seen.
- iii. If a drawing is made with the help of the magnifying lens then the magnification of the drawing in relation to the size of the object **must** be worked out as shown below.

$$\text{Magnification} = \frac{\text{length of the drawing}}{\text{actual length}}$$

Example 1

- A form one student examined a specimen whose length was 43mm, then drew the diagram whose length was 86mm. Calculate the magnification of the drawing.

Solution.

$$\begin{aligned} \text{Magnification} &= \frac{\text{length of the drawing}}{\text{length of the object}} \\ &= \frac{86 \text{ mm}}{43 \text{ mm}} \\ &= \times 2. \end{aligned}$$

Example 2

- If the magnification of a drawing is x5 and the drawing length is 10cm. What is the actual length of the object?

solution

$$\text{Magnification} = \frac{\text{length of the drawing}}{\text{actual length}}$$

$$\text{Actual length} = \frac{\text{length of the drawing}}{\text{magnification}}$$

$$\text{Actual length} = \frac{10\text{cm}}{5}$$

$$\text{Actual length} = 2\text{cm}$$

BIOLOGICAL DRAWINGS

- The following are important points to note when making biological diagrams:
 1. Use a well sharpened pencil.
 2. The drawing should occupy $\frac{1}{2}$ or $\frac{3}{4}$ of the space provided.
 3. Each drawing should have a title.
 4. Enough space should be left all round the drawing for labeling.
 5. Avoid using double lines when making outlines of a drawing.
 6. Label lines should not have arrow heads.
 7. The magnification of a drawing should always be worked out.
 8. The drawing should not be shaded.
 9. The label lines should never cross each other.

Taxonomic Units of Classification.

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- These refer to the groups (or taxa) into which organisms are placed.
- There are seven major taxonomic units as shown below.
 - i. Kingdom.
 - ii. Phylum (or Division).
 - iii. Class.
 - iv. Order.
 - v. Family.
 - vi. Genus.
 - vii. Species.
- Moving down the taxonomic units the number of organisms in each group decreases but the similarities between then increases.
- Phylum is used when classifying animals while division is used when classifying plants.
- The **kingdom** has the highest number of organisms/members while **species** has members with more common characteristics.
- The species is the smallest unit of classification **whose members can naturally and freely interbreed and produce/give rise to fertile offspring.**

KINGDOMS OF CLASSIFICATION

1. **Monera**-This includes the bacteria and viruses.
2. **Protocista**- This includes the algae, e.g. spirogyra and protozoa, e.g. amoeba, Paramecium and Plasmodium.
3. **Fungi**- This includes moulds, e.g. moulds, yeasts and mushrooms.
4. **Plantae**- Examples include moss, blacken fern, maize, beans and jacaranda.
5. **Animalia**- This includes housefly, spider, crab, lizard, elephant, hawk and cow.

SCIENTIFIC NAMING OF ORGANISMS.

- The present system of naming organisms is called **Binomial nomenclature**.
- It was developed by Swedish biologist called ***Carolus Linnaeus*** in 18th century.
- It involved giving organisms two latin names because:
 - i. They rarely change/ are static.
 - ii. They are written in the same language all over the world.

Importance of giving organisms two names

- i. Enables biologists to arrange organisms in an orderly manner.
- ii. It provides names that have the same meaning world wide.

BINOMIAL NOMENCLATURE

- This is the scientific system of giving organisms two names, generic/ genus name and specific/ species name.

Rules of binomial nomenclature.

1. The first name/ genus name should begin with capital letter and the second name/ specific name should be written in small letters.
2. The names should be printed in italics in books but underlined separately when handwritten.

Example,

- The lion is called *Panthera leo* when printed and Panthera leo when hand written the leopard is called *Panthera pardus* when typed and Panthera pardus when hand written.
- The name *Panthera* represents genus while *leo* and *pardus* represent species.
- Lion and leopard belong to the same genus but different species.